

26 Importance of E101

A signal is a mathematical abstraction for any measurement.

All engineering devices interact with signals: voltage, current, temperature, force, altitude, pressure, concentration, image, ekg, mri, audio recording, etc. Signals.

Signals in the real world are continuous, (CTFT and Fourier Series come in to play) but our representations are discrete (DTFT, DFT).

Developing an engineering device requires understanding continuous and discrete time signals.

Modeling systems

A system is a mathematical abstraction for input-output behavior. We don't care about the details - just how the input is acted upon to make the output.

LTI systems are simple and general, and apply to any domain of engineering.

Applications

Compression:

A lot of media is stored in compressed formats.

audio: mp3, m4a, flac.

image: jpeg, tiff.

video: mp4, avi, webm.

All rely on Fourier analysis.

EG: mp3 → (an efficient implementation of the discrete Fourier series)

Convolutional Neural Networks

combine convolution & neural network

Huge role in the AI Revolution

Alexnet